

Lab Work : Python Selection Statements

1. Smart Income Tax Calculator

Problem Statement

A government tax portal calculates tax based on the following conditions:

- Income up to ₹5,00,000 → No tax
- ₹5,00,001 to ₹10,00,000 → 10%
- ₹10,00,001 to ₹20,00,000 → 20%
- Above ₹20,00,000 → 30%

Additional Benefits:

- Senior citizens (Age ≥ 60) receive a 5% rebate on tax.
- Women taxpayers receive an additional 2% rebate on tax.

Write a Python program to calculate the final tax amount payable.

Sample Input

Enter Annual Income: 1200000

Enter Age: 65

Enter Gender (M/F): F

Sample Output

Tax before rebate: ₹240000.0

Senior Citizen Rebate: ₹12000.0

Women Rebate: ₹4800.0

Final Tax Payable: ₹223200.0

2. Loan Approval System

Problem Statement

A bank evaluates loan applications using:

- Credit Score ≥ 750
- Annual Income $\geq ₹8,00,000$
- Existing Loan Amount $\leq ₹2,00,000$

Conditions:

- All conditions satisfied → Approved
- Only one condition fails → Manual Review

- More than one condition fails → Rejected

Sample Input

Enter Credit Score: 780

Enter Annual Income: 750000

Enter Existing Loan Amount: 100000

Sample Output

Loan Status: Manual Review

Reason: Income criteria not satisfied.

3. Airline Ticket Pricing Engine

Problem Statement

An airline calculates ticket fare using:

Base Fare = ₹5000

Additional Charges:

- Business Class → +₹3000
- Window Seat → +₹500
- Weekend Travel → +₹1000

Discounts:

- Age below 12 → 50%
- Age above 60 → 20%

Calculate the final ticket fare.

Sample Input

Enter Passenger Age: 65

Business Class (Y/N): Y

Window Seat (Y/N): Y

Weekend Travel (Y/N): Y

Sample Output

Base Fare: ₹5000

Additional Charges: ₹4500

Senior Citizen Discount: 20%

Final Ticket Fare: ₹7600.0

4. Employee Performance Evaluation

Problem Statement

An employee is evaluated using:

- Project Score
- Attendance Percentage
- Client Feedback Score

Rules:

- Excellent → All scores above 90
- Good → Scores above 75
- Average → Scores above 60
- Poor → Otherwise

Additional Rule:

- Attendance below 70% cannot receive more than Average rating.

Sample Input

Project Score: 95

Attendance: 65

Client Feedback: 92

Sample Output

Performance Rating: Average

Reason: Attendance below 70%

5. E-Commerce Premium Membership Eligibility

Problem Statement

A customer becomes Premium Member if:

- Total Purchases > ₹50,000
- Orders Completed $\geq$ 20
- Customer Rating $\geq$ 4.5

Special Case:

- Purchases above ₹1,00,000 automatically qualify.

Sample Input

Total Purchases: 120000

Orders Completed: 10

Customer Rating: 4.0

Sample Output

Premium Membership Status: Eligible
Reason: Purchase amount exceeded ₹100000.

6. University Scholarship System

Problem Statement

Scholarship is awarded based on percentage:

Percentage	Scholarship
95+	100%
90-94	75%
85-89	50%
80-84	25%
Below 80	No Scholarship

Conditions:

- Family income must be below ₹8,00,000.
- Students with disciplinary actions receive no scholarship.

Sample Input

Percentage: 92
Family Income: 500000
Disciplinary Action (Y/N): N

Sample Output

Scholarship Awarded: 75%

7. Cybersecurity Login Validation

Problem Statement

A login system validates:

- Username
- Password
- OTP

Conditions:

- All correct → Login Successful
- Incorrect OTP → Re-enter OTP
- Incorrect Password after 3 attempts → Account Locked

- Incorrect Username → User Not Found

Sample Input

Username: admin

Password: admin123

OTP: 4567

Sample Output

Login Successful

Welcome Admin

8. Smart Electricity Billing System

Problem Statement

Calculate electricity bill using:

Units	Rate
0-100	₹5/unit
101-300	₹7/unit
Above 300	₹10/unit

Additional Rules:

- Commercial consumers pay 20% extra.
- Bills above ₹5000 attract 5% surcharge.
- Senior citizens receive 10% discount.

Sample Input

Units Consumed: 450

Consumer Type (Residential/Commercial): Commercial

Senior Citizen (Y/N): Y

Sample Output

Base Bill: ₹4500

Commercial Charge: ₹900

Surcharge: ₹270

Senior Citizen Discount: ₹567

Final Bill Amount: ₹5103

9. Online Examination Result Analyzer

Problem Statement

A student appears in 5 subjects.

Rules:

- Minimum 40 marks in every subject to pass.
- Distinction $\rightarrow$ Average ≥ 75
- First Division $\rightarrow$ Average ≥ 60
- Second Division $\rightarrow$ Average ≥ 50
- Pass $\rightarrow$ Average ≥ 40
- Fail if any subject score < 40

Sample Input

Hindi : 85

English : 78

Mathematics : 82

Science : 75

Computer : 90

Sample Output

Average Marks: 82.0

Result: PASS

Classification: Distinction

10. Multi-Level Access Control System

Problem Statement

Assign access levels based on:

Roles:

- Admin
- Manager
- Employee
- Guest

Conditions:

- Admin + Security Clearance $\geq 5 \rightarrow$ Full Access
- Manager + Experience > 5 Years $\rightarrow$ Department Access
- Employee + Experience > 2 Years $\rightarrow$ Limited Access
- Guest $\rightarrow$ Read-Only Access
- Inactive Account $\rightarrow$ No Access

Sample Input

Role: Admin

Security Clearance: 6

Account Status: Active

Sample Output

Access Level: FULL ACCESS

Bonus Challenge – Hospital Emergency Triage System

Problem Statement

A hospital prioritizes patients based on:

- Critical Condition
- Age
- Oxygen Level

Rules:

- Critical condition → Immediate Treatment
- Oxygen below 90 → High Priority
- Age above 65 → Medium Priority
- Others → Normal Priority

Sample Input

Critical Condition (Y/N): N

Age: 70

Oxygen Level: 94

Sample Output

Patient Priority: Medium Priority

Reason: Senior Citizen